

Gut Microbiome Characterization and Machine Learning-Based Prediction of Type 2 Diabetes in Urban African Adults

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EXECUTIVE SUMMARY

Background: Type 2 diabetes (T2D) is a public health concern globally, with a rapidly increasing burden in Africa. Recent studies suggest that alterations in the gut microbiome contribute to T2D pathogenesis through effects on glucose metabolism, inflammation, and immune regulation. Machine learning (ML) offers powerful approaches for analyzing high-dimensional microbiome data and identifying microbial signatures associated with disease. However, most microbiome-based prediction models have been developed using non-African populations, limiting their applicability to African settings. This study aimed to characterize gut microbiome alterations associated with T2D and evaluate the performance of multiple machine learning algorithms for predicting T2D among urban African adults using gut microbial profiles.

Methodology: A secondary analysis was conducted using publicly available 16S rRNA sequencing data from 131 urban African adults (66 T2D cases and 65 healthy controls) from Ibadan, Nigeria. Bioinformatics processing included sequence quality control, denoising, chimera removal, taxonomic assignment, and generation of amplicon sequence variants (ASVs). Microbial diversity was assessed using the Shannon diversity index, Bray-Curtis dissimilarity, principal coordinates analysis (PCoA), Permutational Multivariate Analysis of Variance (PERMANOVA), taxonomic composition analysis, and Analysis of Compositions of Microbiomes with Bias Correction 2 (ANCOM-BC2) differential abundance analysis. Microbial abundance data were normalized and filtered before developing Random Forest, Support Vector Machine (SVM), and Logistic Regression models. Model performance was evaluated using accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (AUROC).

Results: Alpha diversity did not differ significantly between healthy individuals and participants with T2D (Shannon index, $p = 0.19$). Although microbial communities showed considerable overlap, beta diversity analysis demonstrated significant compositional differences between groups (PERMANOVA: $F = 1.65$, $R^2 = 0.013$, $p = 0.026$). Firmicutes was the dominant phylum in both groups, while *Blautia*, *Clostridium sensu stricto* 1, *Collinsella*, and *Romboutsia* were the predominant genera. Differential abundance analysis identified 19 genera enriched in T2D and two genera enriched in healthy controls, suggesting depletion of beneficial short-chain fatty acid (SCFA) associated bacteria and enrichment of metabolically relevant taxa in T2D. Among the machine learning models, Random Forest demonstrated the best predictive performance, outperforming Support Vector Machine and Logistic Regression across classification metrics, with feature importance analysis identifying several microbial taxa that contributed substantially to disease prediction.

Conclusion and Recommendations: Gut microbiome composition in urban African adults with T2D is characterized by subtle but significant alterations in metabolically important bacterial taxa despite preservation of overall microbial diversity. Machine learning, particularly the Random Forest algorithm, effectively distinguished individuals with T2D from healthy controls using gut microbial profiles, highlighting the potential of microbiome-based artificial intelligence approaches for disease prediction. Future studies should validate these findings in larger African cohorts and integrate metagenomic, metabolomic, and longitudinal data to improve biomarker discovery and enhance the clinical translation of microbiome-based T2D prediction models.

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ABBREVIATION AND ACRONYM

CHAPTER 1: INTRODUCTION

1.1 Background

The gut microbiome has emerged as an important regulator of metabolic health, influencing host energy metabolism, inflammation, immune function, and glucose homeostasis. Dysbiosis of gut microbial communities has been associated with metabolic disorders, including Type 2 diabetes mellitus (T2D). In a landmark study among urban Africans in Nigeria, Doumatey et al., (2020) demonstrated that T2D was associated with significant alterations in gut microbial composition, identifying specific taxa enriched among individuals with diabetes and others reduced compared with healthy controls. These findings provided evidence that microbiome signatures may contribute to T2D pathogenesis and could serve as potential biomarkers for disease detection (Giuffré et al., 2023; Neri-Rosario et al., 2023).

However, gut microbiome composition varies substantially between populations due to differences in geography, diet, lifestyle, genetics, and environmental exposures. Notably, Kc & Paudel, (2025) highlighted significant microbiome variation among African American populations and emphasized the importance of inclusive microbiome datasets for developing equitable disease prediction models. Despite increasing evidence, African populations remain underrepresented in microbiome research, limiting the applicability of models developed in other populations.

Additionally, advances in high-throughput sequencing technologies have generated large-scale microbiome datasets containing thousands of microbial features. Machine learning provides suitable analytical approaches for these complex datasets by identifying nonlinear relationships, selecting important microbial features, and developing predictive models. Similarly, Li et al., (2022) emphasized the growing application of machine learning for integrating human gut microbiome data, while Giuffré et al., (2023) described machine learning as an emerging approach for improving understanding of microbiome disease relationships. Further, recent reviews have underscored the increasing use of explainable artificial intelligence approaches to improve transparency and clinical applicability of microbiome based models (Tourab et al., 2025).

Previous studies have demonstrated the potential of machine learning approaches for predicting T2D using gut microbiome profiles. For example, NavarraBiomed et al., (2021) applied multiple feature selection approaches combined with machine learning classifiers to identify potential microbial biomarkers of T2D, demonstrating that a limited number of microbial taxa could provide meaningful predictive information. Likewise, Neri-Rosario et al., (2023) compared machine learning algorithms in a Mexican cohort and reported that Random Forest achieved strong predictive performance in distinguishing individuals with T2D from controls. More recently, Caballero-María et al., (2025) developed a deep learning model integrating clinical, biochemical, and gut microbiota characteristics to predict T2D risk, demonstrating the expanding potential of artificial intelligence approaches in metabolic disease prediction.

Although previous studies have demonstrated that gut microbiome profiles contain predictive information for T2D, most machine learning models have been developed using non-African populations. Differences in

microbial composition across populations may limit the transferability of existing models. Furthermore, limited studies have evaluated and compared multiple machine learning algorithms for T2D prediction using gut microbiome profiles among African populations. Therefore, this study aimed to develop, evaluate, and compare the performance of multiple machine learning models for predicting Type 2 diabetes (T2D) using gut microbiome profiles among urban African adults.

1.2 Problem Statement

Type 2 diabetes (T2D) is a growing public health challenge globally and in Africa. Nonetheless, previous studies from Europe, North America, China, and animal models have demonstrated associations between gut microbiome composition and T2D, African populations remain underrepresented in microbiome research. The gut microbiome is influenced by geography, diet, genetics, lifestyle, and environmental exposures, making it inappropriate to directly generalize findings and predictive models developed in non-African populations to African settings. Consequently, there is limited understanding of the gut microbial signatures associated with T2D among urban African adults and insufficient evidence on their utility for disease prediction. Addressing this knowledge gap is essential for identifying population-specific microbial biomarkers and developing accurate machine learning models that can support early detection and precision management of T2D in African populations.

1.3 The General and Specific Objectives

1.3.1 General Objective

To characterize gut microbiome profiles and develop a machine learning model for predicting Type 2 diabetes among urban African adults.

1.3.2 Specific Objectives

1. Characterize alpha and beta diversity patterns between individuals with Type 2 Diabetes (T2D) and healthy controls.
2. Compare gut microbial composition between T2D and healthy controls.
3. Identify the microbial taxa that best discriminate between T2D and healthy controls and assess their biological plausibility.
4. Develop and evaluate a classifier that predicts disease status (T2D or healthy) from taxonomic abundance profiles.

CHAPTER II: METHODOLOGY

2.1 Study Design

This study employed a secondary data analysis design using existing gut microbiome sequencing data from urban African adults. The dataset was originally generated by Doumatey et al., (2020) to investigate associations between gut microbiota composition and Type 2 diabetes (T2D) among urban Nigerian adults in Ibadan. In the present study, the same microbiome dataset was reanalyzed using machine learning approaches to develop and evaluate predictive models for classifying individuals with T2D and healthy controls based on gut microbial profiles.

The study applied supervised machine learning algorithms to identify microbial features associated with T2D and assess their ability to predict disease status. Multiple classification models, including Random Forest, Support Vector Machine (SVM), Gradient Boosting and Logistic Regression, were developed and compared based on predictive performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, and receiver operating characteristic - Area under the curve (ROC-AUC).

2.2 Study Population and Data Source

The study utilized a part of 290 publicly available gut microbiome sequence data from a study conducted in Ibadan, Nigeria, an urban area. The analysis used a subset of 131 samples with 66 T2D cases and 65 healthy controls. Participants were part of a primary case-control study (Doumatey et al., 2020). Diabetes was defined as fasting plasma glucose (FPG) ≥ 126 mg/dl or a 2-hour post-load value ≥ 200 mg/dl, and the controls were required to have an FPG < 126 mg/dl and no symptoms of diabetes. Fecal samples were collected during clinical visits and frozen (-80°C). DNA extraction was performed using MoBioPowerMag[®]R Microbiome kit and quantification was done using Qubit[®]R Quant-iT dsDNA High Sensitivity Kit to ensure sufficient DNA concentration. The study targeted the bacterial 16S V4 rRNA region. Sequencing was performed on Illumina Miseq platform after library preparation (paired-end sequencing).

2.3 Data Retrieval

A comma separated value (csv) file with information on microbiome studies done in Africa was downloaded from the African Human Microbiome Portal (<https://microbiome.h3abionet.org/>). We searched for a study with a case-control design and it investigated the association of gut microbiome with a disease and used an amplicon-based approach for sequencing. A study that met the criteria mentioned was selected, Doumatey et al., (2020). The sampleID were extracted from the csv file into a text file and used to retrieve raw fastq sequence reads from SRA-NCBI.

2.4 Bioinformatics analysis

Low-quality sequence reads ($Q < 20$) were trimmed using FASTP and quality checked using FASTQC and MULTIQC. The DADA2 library in R v4.6.1 was used to denoise and dereplicate the trimmed fastq files. The paired-end reads were merged followed by generation of amplicon sequence variant (ASV) tables. Chimeras were also removed using the DADA2 package, taxonomic units at the genus level were assigned using SILVA

Database (silva_nr99_v138.1_train_set.fa.gz). A phyloseq object was created using the phyloseq library in R by merging the ASVs table, the ASV taxonomic tables and the metadata. Unwanted taxa, family mitochondria and order chloroplast, were removed. Genera in less than 10% of the samples were also removed.

2.5 Downstream analyses

Downstream analyses were performed using the R software. The following packages were used for the analysis and visualization; phyloseq, ggplot2, vegan, and ANCOMBC, svglite.

2.5.1 Alpha diversity

To explore variations within the samples, alpha diversity was performed. The Shannon diversity index was used to measure the alpha diversity. Box- and- whiskers plots with overlaid scatter points were used to visualize the distribution of the Shannon diversity index across all samples and between the T2D and healthy control groups. The median and interquartile range (IQR) were used to summarize the Shannon diversity index for each group. Statistical significance between groups was assessed using the Wilcoxon rank-sum test.

2.5.2 Beta diversity

Principal coordinate analysis (PCoA) based on the Bray-Curtis dissimilarity index was used to visualize differences in microbial community composition. Statistical significance was assessed using permutational multivariate analysis of variance (PERMANOVA).

2.5.3 Microbial composition

The mean relative abundance of bacterial phyla and genera were compared between the T2D and healthy control groups using stacked bar charts. The 15 most abundant genera were further visualized using side-by-side bar charts to facilitate comparison of their mean relative abundance between the two groups.

2.5.4 Differential abundance analysis

To identify genera that are most discriminative of the two groups, a differential abundance analysis was performed using Analysis of Composition of Microbiomes with Bias Correction (ANCOM-BC). The Benjamini-Hochberg procedure was used to correct for multiple testing. The T2D group was used as the comparison group and the healthy controls as the reference group.

2.6 Data Preprocessing for Machine Learning

Microbial abundance data were transformed into a feature matrix suitable for machine learning analysis, where rows represented individual participants and columns represented microbial taxa. Each feature represented the relative abundance of a microbial taxon, while the corresponding disease status (T2D or healthy control) was used as the outcome variable for model training and evaluation.

Normalization

Taxonomic abundance data were normalized using centered log-ratio transformation to account for compositional characteristics of microbiome data.

Feature filtering

Low prevalence microbial features were removed to reduce dimensionality and improve model stability.

Handling class imbalance

2.6 Machine Learning Model Development

2.6.1 Classification algorithms

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve classification accuracy and identify important microbial predictors.

Support Vector Machine (SVM)

SVM identifies an optimal decision boundary separating T2D and control groups and is effective for high-dimensional datasets.

Logistic Regression

Logistic regression was included as an interpretable baseline classifier for binary disease prediction.

2.7 Model Training and Validation

The dataset was divided into training and testing subsets. Models were trained using the training dataset and evaluated on unseen test data to estimate predictive performance.

2.8 Model Performance Evaluation

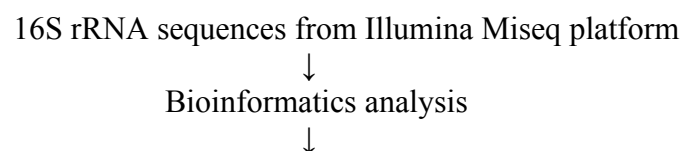
2.9 Feature Importance and Biomarker Identification

Important microbial features contributing to T2D prediction were identified using feature importance scores from machine learning models. These taxa were compared with previous literature to assess biological relevance.

2.10 Statistical Analysis Software

Statistical analyses were performed using R version 4.6.1 and Python version 3.12.13. Microbiome analyses were conducted using the R packages phyloseq and microbiome, while machine learning models were developed using the Python scikit-learn library.

Conceptual workflow diagram



Taxonomic abundance table



Filtering + normalization



Feature selection



Machine learning models

- Random Forest
- SVM
- Logistic Regression
- GradientBoosting



Performance evaluation

- Accuracy
- Sensitivity
- Specificity
- F1-score
- AUROC



Important microbial predictors

CHAPTER III: RESULTS AND DISCUSSION

3.1 Results

Alpha diversity, as measured by the Shannon diversity index, did not differ significantly between healthy individuals and participants with type 2 diabetes (T2D). The median Shannon diversity was 3.496 (IQR: 3.249–3.771) in the healthy group and 3.566 (IQR: 3.346–3.879) in the T2D group. Although the T2D group exhibited a slightly higher median Shannon diversity, the difference was not statistically significant ($p = 0.19$) (Figure 3.1).

Principal coordinates analysis (PCoA) of Bray–Curtis distances revealed substantial overlap in the gut microbiota composition of healthy and T2D participants, with no distinct clustering observed between the groups (Figure 3.2). Nevertheless, PERMANOVA indicated that microbial community composition differed significantly between the two groups ($F = 1.65$, $R^2 = 0.013$, $p = 0.026$). Disease status accounted for 1.3% of the total variation in microbial community structure.

At the phylum level, Firmicutes constituted the majority with percent relative abundance of 78% followed by Actinobacteriota (mean: 16.5%) and Euryarchaeota (mean: 2.2%) (Figure 3), which aligns with the typical gut microbiome structure in adults (Doumately et al., (2020).

At the genus level, both healthy and T2D groups were dominated by members of *Blautia* as the most abundant genus followed by *Clostridium sensu stricto 1*, *Collinsella* and *Romboutsia*. Other genera such as *Ruminococcus*, *torques group*, *Streptococcus* and *Turicibacter* were identified with an abundance of less than 5%. Visual inspection revealed differences in the relative abundance between the two groups. The proportion of rare taxa grouped as ‘Others’ was higher in T2D individuals compared to healthy controls. Additionally, Genera such as *Blautia*, *Clostridium sensu stricto 1* and *Bifidobacterium* appeared to be more abundant in healthy individuals (Figure 3.4 and 3.5)

The ANCOM-BC2 analysis revealed that at a false discovery rate threshold of 0.05, 19 genera were significantly enriched in individuals with T2D, while 2 genera were significantly depleted (Figure 5). Genera most strongly associated with T2D included *innocuum group*, *Rikenellaceae RC9 gut group*, *Dialister*, *Succinivibrio* and *Limosilactobacillus*. In contrast, ventriosum group and Romboutsia were significantly more abundant in healthy controls. These results indicate a shift toward increased abundance of several genera and a reduction in butyrate-producing and health-associated taxa in T2D.

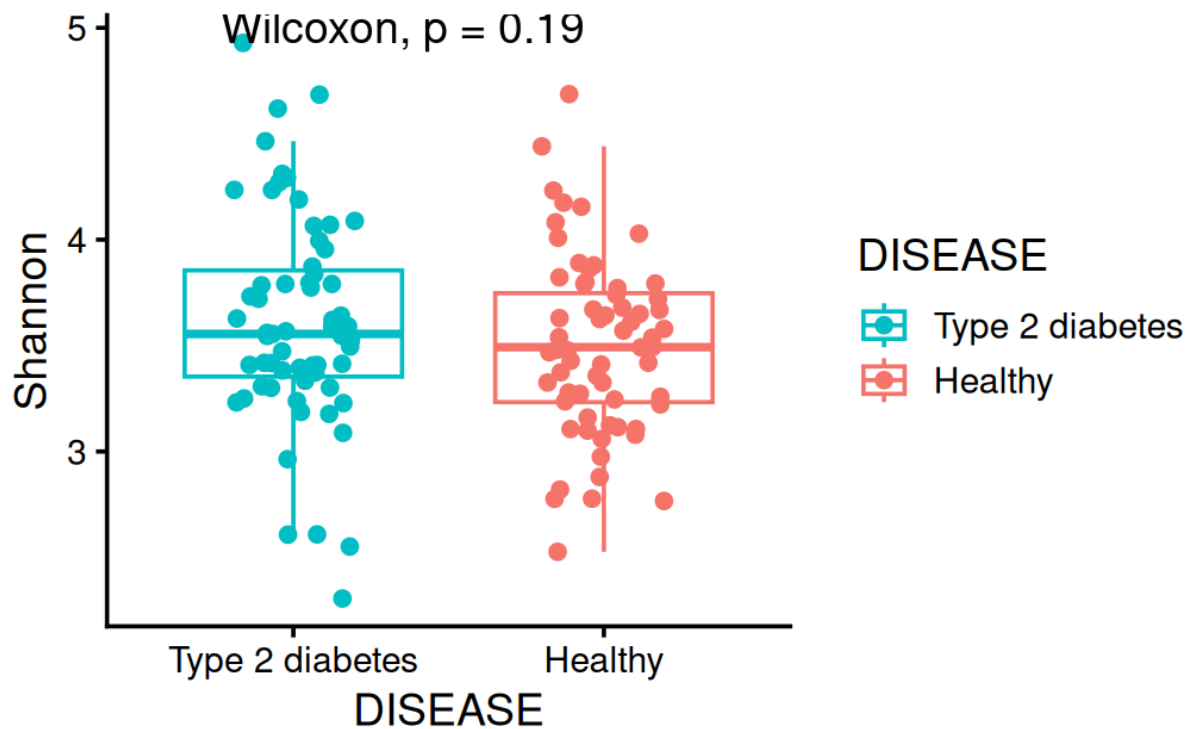


Figure 3.1. Shannon alpha diversity index within samples of T2D patients and healthy individuals.

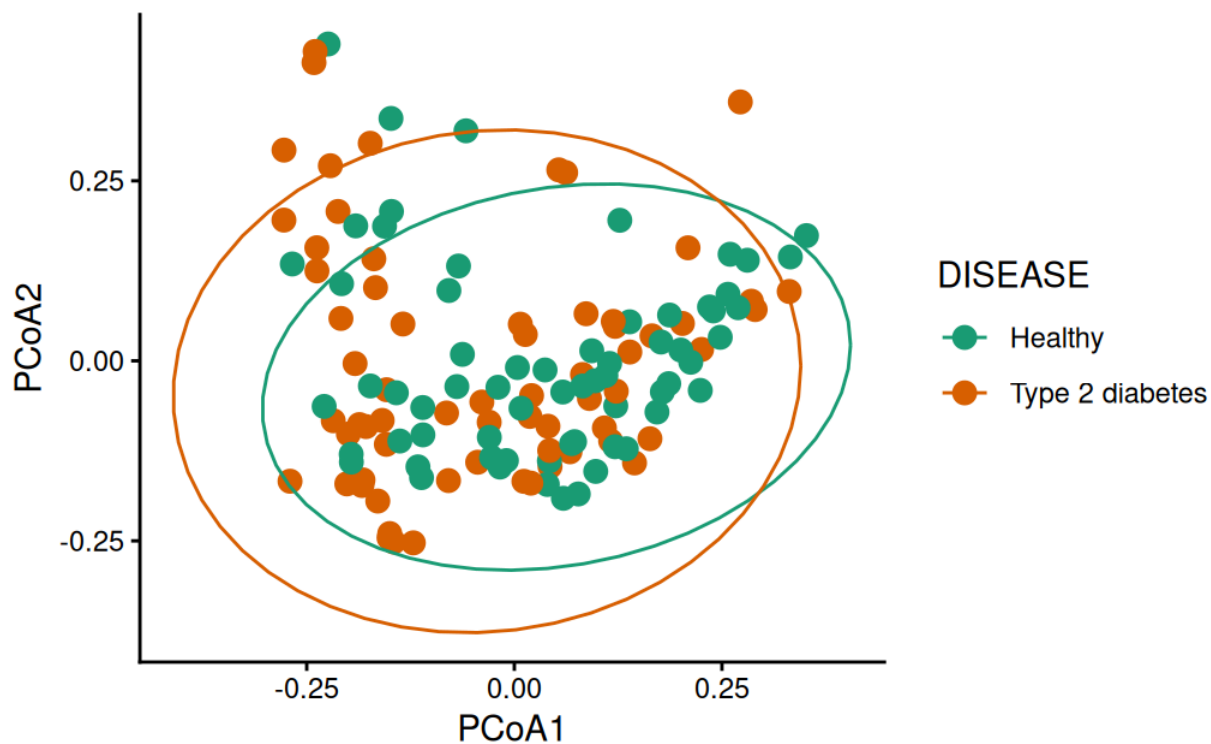


Figure 3.2. Beta diversity of gut microbiota composition of healthy and T2D participants

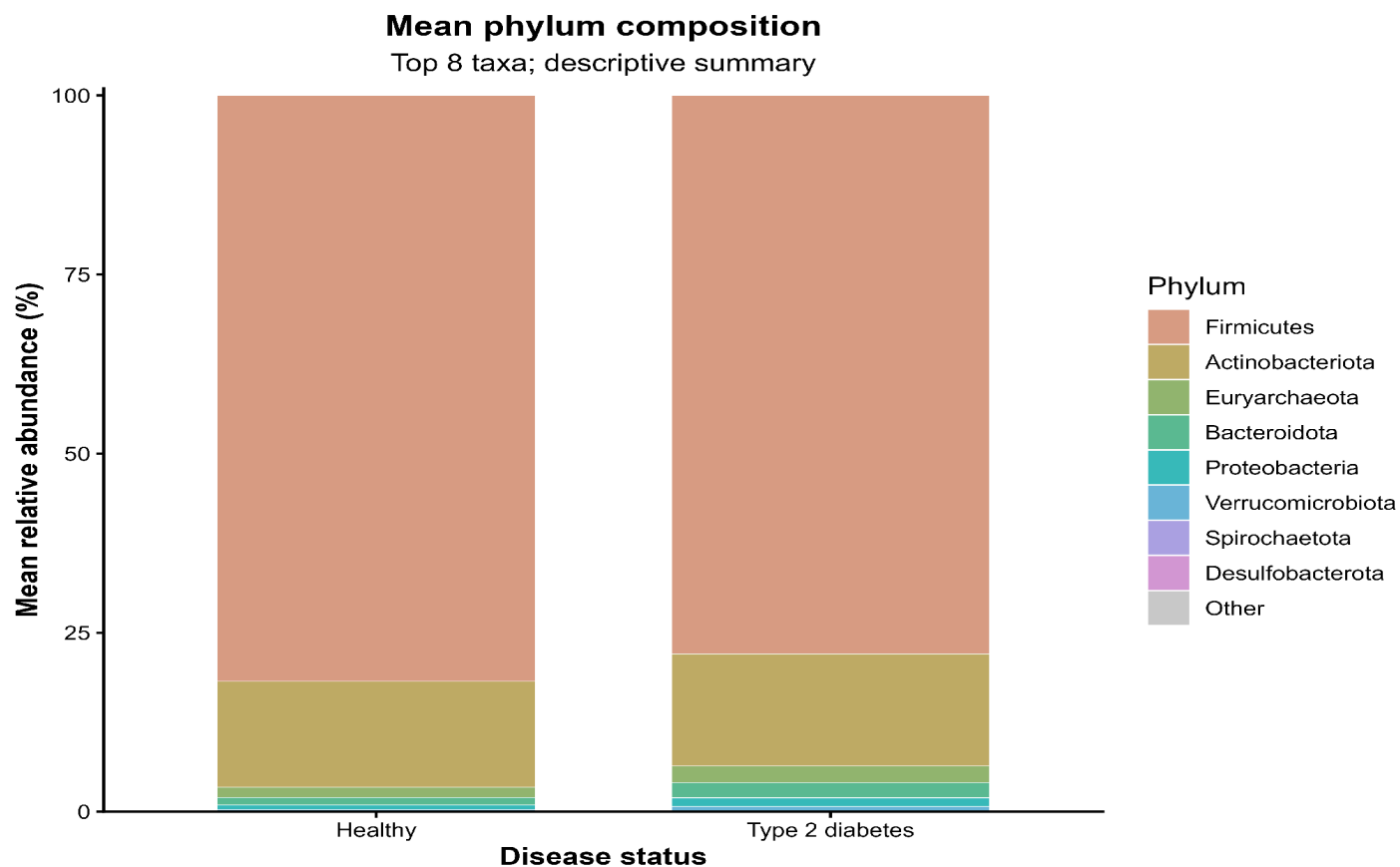


Figure 3.3. Mean phylum composition of the gut microbiome in healthy controls and individuals with T2D. Bars represent mean relative abundance (%) of top 8 bacterial phyla.

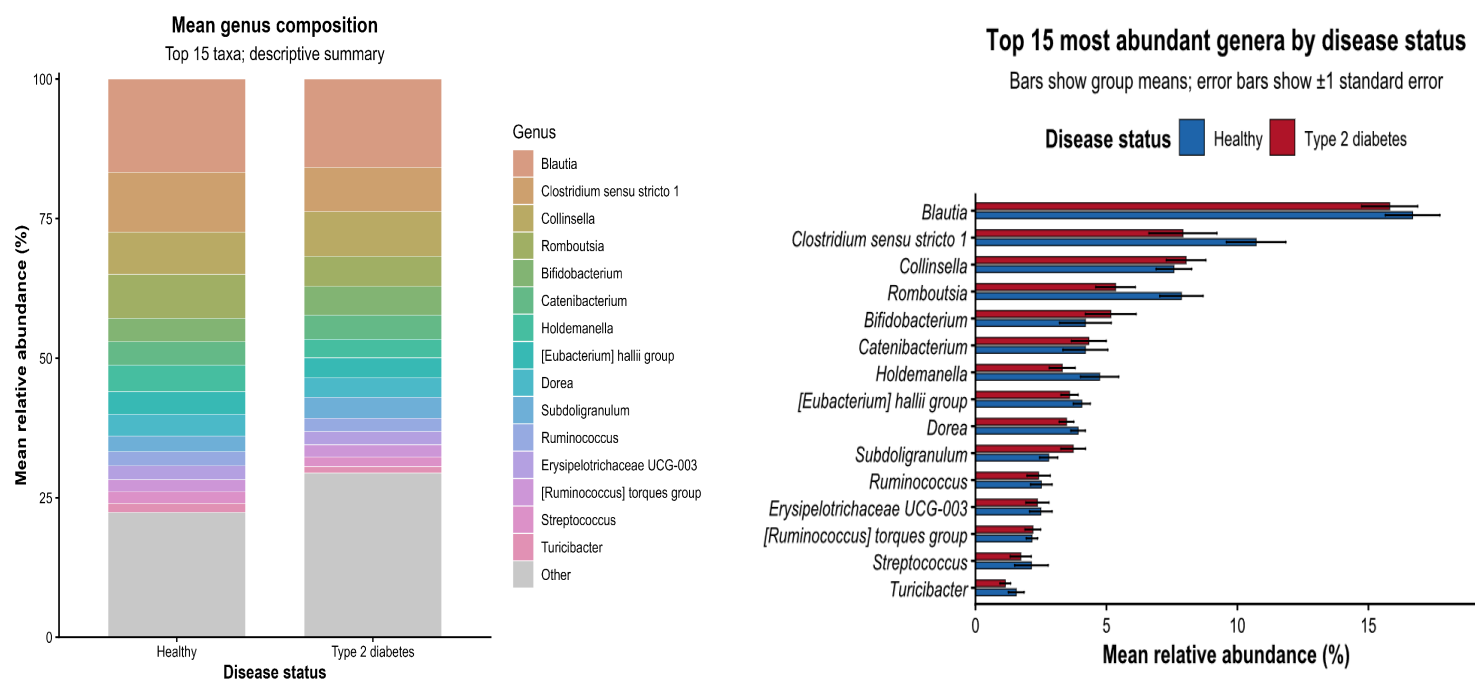


Figure 3.4. Mean genus composition of the gut microbiome in healthy controls and individuals with T2D.

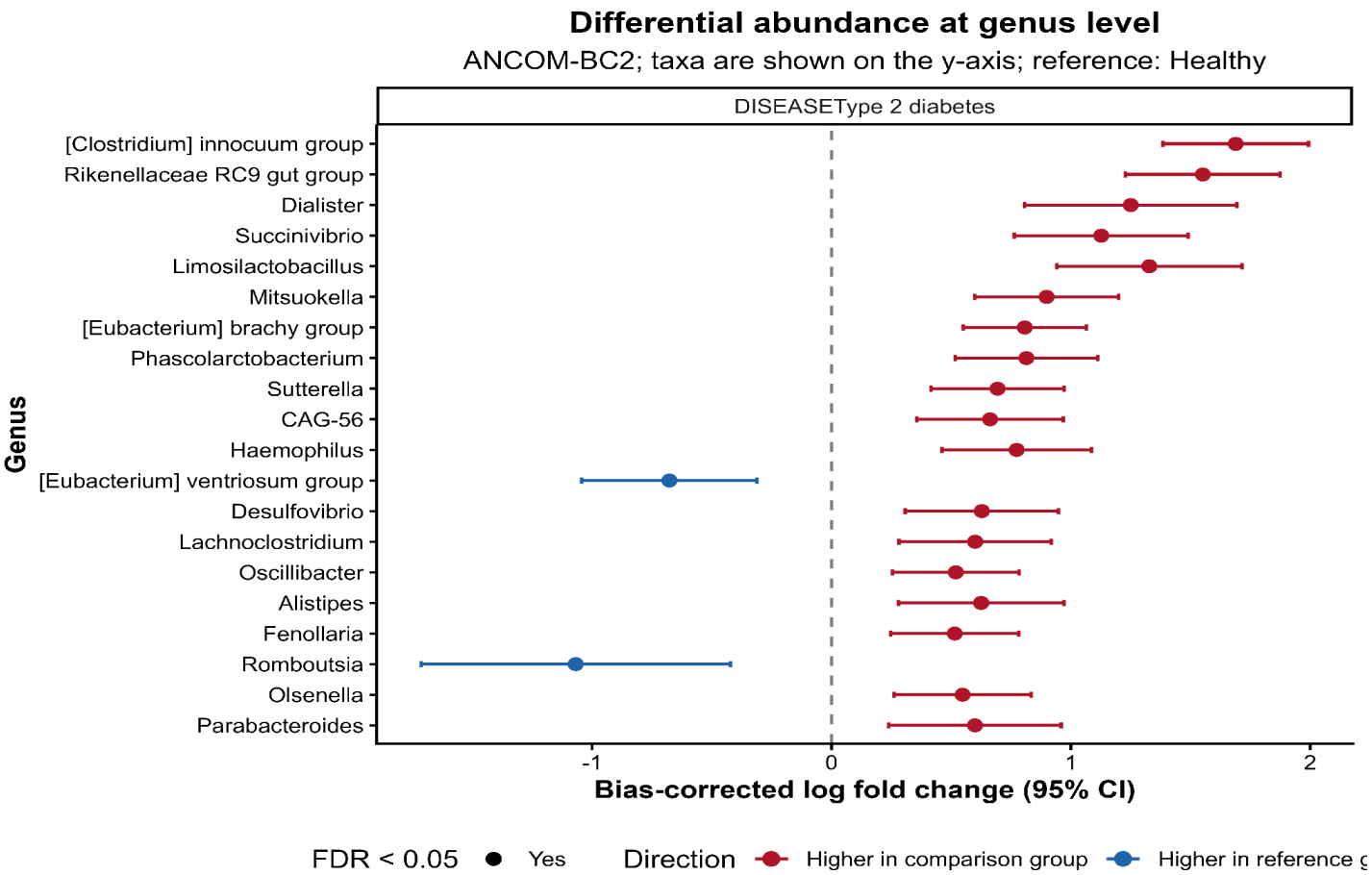


Figure 3.5. Forest plot of bias-corrected log fold change (95% CI) of differentially abundant genera between T2D and healthy controls. Red = higher in T2D, Blue = higher in Healthy. FDR < 0.05.

CHAPTER IV: DISCUSSION

In this study, we characterized gut microbial diversity and composition in urban African adults with T2D compared to healthy controls. Our findings suggest that while diversity remains largely stable, specific taxonomic shifts occur in T2D and may contribute to disease pathophysiology.

We found no significant difference in alpha diversity between T2D and healthy individuals, as measured by the Shannon index ($p = 0.19$). This is consistent with several studies in African and Asian populations where T2D is not always associated with reduced within-sample diversity (Karlsson et al., 2013). However, some Western cohorts have reported decreased diversity in T2D, suggesting that geography, diet, and host genetics may modulate this relationship (J. Wang et al., 2012).

Beta diversity analysis using Bray-Curtis distances showed substantial overlap between groups, although PERMANOVA identified a significant difference in microbial community structure, with disease status explaining a small proportion of the variation ($R^2 = 0.013$, $p = 0.026$). This indicates that although the overall microbiome structure is similar, T2D is associated with subtle but detectable compositional changes. The low R^2 suggests that other factors such as diet, medication, and host genetics likely play larger roles in shaping the microbiome (L. Wang et al., 2021).

The persistence of dominant genera across healthy and T2D groups suggests that the overall gut microbial community structure remained relatively conserved, with disease-associated changes occurring primarily among specific metabolically relevant taxa rather than through extensive restructuring of the microbial ecosystem. This observation is consistent with previous studies indicating that T2D is associated with selective alterations in microbial composition and function, particularly involving taxa linked to glucose metabolism and microbial metabolite production rather than uniform loss of microbial diversity (Liu et al., 2022; Palmnäs-Bédard et al., 2022).

The observed reduction of short-chain fatty acids (SCFA) -associated bacteria is biologically plausible given the established role of microbial-derived SCFAs, including acetate, propionate, and butyrate, in maintaining intestinal barrier integrity, regulating immune responses, modulating inflammation, and influencing glucose homeostasis (Salamone et al., 2021; Portincasa et al., 2022). Recent study further highlights that SCFAs act as signaling molecules that participate in metabolic regulation, insulin sensitivity, and host-microbiota communication, including interactions involving neuroendocrine and immune pathways (Xie et al., 2025). Therefore, reductions in SCFA-associated taxa observed in the T2D group may indicate altered microbial functional potential that could contribute to impaired metabolic regulation. However, direct confirmation would require metagenomic, metatranscriptomic, or metabolomic analyses measuring SCFA production.

Alterations involving genera such as *Blautia* and *Bifidobacterium* further support the possibility of ecosystem-level functional changes. Members of these groups are involved in carbohydrate fermentation, metabolite exchange, and microbial cross-feeding networks that contribute to gut microbial stability. Previous studies have reported associations between *Bifidobacterium* and improved metabolic profiles, while changes in *Blautia* abundance have been observed in relation to metabolic disorders, although findings vary depending on population characteristics, diet, and methodological approaches (Palmnäs-Bédard et al., 2022; Liu et al., 2022). [GK3] Thus, the observed shifts in these taxa may reflect altered microbial interactions associated with T2D rather than a generalized disruption of the gut microbiome.

The enrichment of T2D-associated taxa alongside the depletion of beneficial microbial groups suggests a shift toward a dysbiotic gut environment characterized by altered metabolic potential. Increased abundance of taxa linked to inflammation and metabolic dysfunction may influence host metabolism through pathways involving inflammatory signaling and bile acid regulation, which are important in glucose homeostasis and insulin sensitivity (Gurung et al., 2020). Conversely, the reduction of butyrate-associated bacteria may impair microbial contributions to intestinal barrier maintenance and immune regulation. Since butyrate supports epithelial integrity and has anti-inflammatory effects, depletion of these functional groups may contribute to the chronic low-grade inflammation commonly observed in T2D (Louis & Flint, 2017).

Machine Learning module

Data Partitioning

The processed data were partitioned into an 80/20 train-test split to ensure model isolation. We utilised stratification (**stratify=y**) to maintain the approximately 50/50 healthy-to-diabetic ratio across both sets. This ensures that the training environment is balanced and the final evaluation is conducted on a representative, unseen biological subset.

3. Exploratory Data Analysis (EDA)

Class Distribution

The cohort is well-balanced for binary classification, with 50.38% of samples (66) representing Type 2 Diabetes and 49.62% (65) representing Healthy controls. This balance mitigates the risk of class-imbalance bias during the learning phase.

Dimensionality Analysis

Principal Component Analysis (PCA) was utilised to map the variance within the microbiome profiles. PC1 explains 13.3% of the variance, while PC2 explains 5.8%. The PCA plot reveals a significant visual overlap between the "Healthy" and "Type 2 diabetes" groups, with only slight separation in the confidence ellipses.

As a senior specialist, I interpret this overlap as evidence that T2D status is not driven by a single dominant biomarker but rather by subtle, non-linear multi-taxa interactions. This justifies the deployment of advanced ensemble models like Random Forest and Gradient Boosting, which can capture these complex dependencies. Key contributors to the observed variance include the genera *Collinsella*, *Blautia*, and *Romboutsia*.

4. Advanced Feature Selection Framework

Methodological Selection

To further refine the model's interpretability and performance, we employed a multi-pronged feature selection framework:

- **Correlation Analysis:** Identification and removal of collinear features.
- **Mann-Whitney U Testing:** Statistical screening for taxa with significant differential abundance between classes.
- **LASSO Regularisation:** Embedded selection using L1 penalties to shrink non-contributory coefficients to zero.
- **Consensus Approach:** An ensemble feature selection method that aggregates the most stable predictors across multiple techniques to enhance model interpretability and robustness.

Feature Taxonomy Examples

Genera identified and processed through this framework include *Bifidobacterium*, *Dorea*, *Clostridium sensu stricto 1*, and *Erysipelotrichaceae UCG-003*.

5. Model Comparison and Benchmarking

Implementation Stack

The modelling pipeline was implemented in Python using the scikit-learn framework. We utilised **StandardScaler** for feature standardisation. Four algorithms were benchmarked: Random Forest (RF), Support Vector Machine (SVM) with **probability=True** enabled for ROC analysis, Logistic Regression (LR), and HistGradientBoosting (GB).

Comparative Performance Table

The following results summarize the performance across different feature subsets and algorithms:

Feature Set	Model			
	Accuracy Precision Recall F1-Score			

All Random
Forest

0.7407 0.8182 0.6429 0.7200

RF50 Random
Forest

0.7037 0.8000 0.5714 0.6667

All Gradient
Boosting

0.7037 0.7500 0.6429 0.6923

Consensus Gradient
Boosting

0.6296 0.7000 0.5000 0.5833

All SVM

0.6296 0.6667 0.5714 0.6154

	Random				
Consensus	Forest				
		0.5926	0.6154	0.5714	0.5926

	SVM				
Consensus					
		0.5556	0.6667	0.2857	0.4000

	Logistic				
LASSO	Regression				
		0.5185	0.5556	0.3571	0.4348

The Random Forest classifier using the full filtered feature set ("All") delivered the highest accuracy (0.7407) and precision (0.8182).

6. Model Evaluation and Diagnostic Metrics

ROC-AUC Analysis

The discriminatory power of the pipeline was assessed using the Area Under the Curve (AUC). The Random Forest Classifier, applied to the "RF10" feature set, achieved an AUC of 0.79, demonstrating strong diagnostic potential for distinguishing T2D from healthy profiles.

Confusion Matrix and Clinical Interpretation

The optimal model's performance on the test set (27 samples) was as follows:

- True Negatives (TN): 10
- True Positives (TP): 8

- **False Positives (FP): 3**

- **False Negatives (FN): 6**

Diagnostic Performance Calculations:

- **Sensitivity (Recall): 57.1%**
- **Specificity: 76.9%**

Clinical Implications: From a clinical perspective, a sensitivity of 57.1% is a significant limitation for a primary screening tool, as nearly half of the diabetic cohort would remain undetected (False Negatives). While the specificity of 76.9% is more robust, reducing the rate of unnecessary follow-up for healthy individuals, the high False Negative rate suggests that the model currently lacks the required "rule-out" capability for definitive clinical application without further optimisation of the recall metrics.

7. Hyperparameter Optimisation and Interpretability

Tuning Strategy

To refine model parameters, we employed **GridSearchCV**. This systematic approach ensured the selection of optimal hyperparameters (e.g., forest depth, estimator counts, and regularisation strengths) tailored to the specific characteristics of microbiome data.

SHAP Analysis

To bridge the gap between black-box machine learning and clinical decision-making, we integrated SHAP (SHapley Additive exPlanations). This provides the "actionable" transparency required by clinicians:

- **Global SHAP:** Identifies which microbial taxa, such as *Blautia* or *Collinsella*, serve as the most significant predictors across the entire population.
- **Local SHAP:** Decomposes individual patient predictions to show exactly how specific increases or decreases in certain taxonomic abundances led the model to a T2D or Healthy classification. This transparency is essential for establishing clinical trust and validating the biological plausibility of the model's findings.

8. Final Deployment and Packaging

Deployment Workflow

The finalised pipeline is packaged as a serialised object containing both the **StandardScaler** for data normalisation and the optimised Random Forest model. This ensures that new metagenomic profiles undergo identical transformations to the training data, preserving predictive validity.

5.1 Conclusion

Overall, our findings suggest that T2D in urban African adults is associated with specific alterations in gut microbial composition rather than a loss of overall diversity. Changes in SCFA-associated and metabolically relevant taxa may reflect microbial shifts linked to inflammation, gut barrier function, and metabolic regulation. Larger longitudinal studies integrating metagenomic and metabolomic approaches are needed to determine whether these microbial changes contribute to T2D development or result from disease-related factors and treatment.

5.2 Strength of the study

This study benefited from the team's established expertise in bioinformatics data analysis and workflow development, which ensured rigorous processing and analysis of the gut microbiome sequencing data. The project was conducted by a collaborative multidisciplinary team that worked effectively toward achieving the study objectives, promoting efficient problem-solving and knowledge sharing throughout the research process. Additionally, the team members were highly motivated and shared a strong commitment to the project's success, contributing to the quality, timely completion, and scientific rigor of the project task.

5.3 Limitation

1. Limited computational infrastructure (server capacity and internet connectivity) necessitated distributing analyses across multiple laptops before merging the outputs.
2. Limited project duration prevented analysis of the full dataset (291 samples); only 131 samples were processed and analyzed.
3. Broad and rapidly evolving literature on the gut microbiome required substantial time to understand global evidence and identify knowledge gaps relevant to type 2 diabetes in African populations.

5.4 Recommendation

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